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RetireOPD

Adaptive Retirement of Teacher Distillation Unlocks an 18-Point Gain for Multi-Turn Agentic RL on ALFWorld and WebShop

By Yan Yu, Zhengxi Lu, Yizhou Liu, Yichen Pan, Aozhe Wang, Qipeng Chen, Hua Yang, Wenqi Zhang, Weiming Lu, Qianglong Chen & Yongliang Shen (Zhejiang University; Alibaba Group)

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Teacher distillation can harm as much as it helps when training agentic reinforcement learning—unless the student knows when to stop listening. *RetireOPD* separates a skill-conditioned teacher from a skill-free student, trains them in sequence, and lets the student autonomously retire distillation once it has absorbed what the teacher can teach. The result: up to +18.8% success rate on ALFWorld and +19.0% on WebShop over pure RL across three model scales.

AT A GLANCE

Problem: Multi-turn agents trained with sparse-reward RL receive weak learning signals; on-policy distillation (OPD) from a privileged teacher helps early but harms later when the teacher becomes a bottleneck.

Why it matters: Web and household agents must make long chains of tool calls; sparse end-of-episode rewards provide almost no signal for individual action improvements.

Method: Pre-train a skill-conditioned teacher with GRPO; jointly train a skill-free student with RL + OPD; retire OPD automatically once the student’s discrepancy stops shrinking and it reaches a target success fraction of the teacher.

Result: +14.1–18.8% on ALFWorld; +11.8–19.0% on WebShop vs. RL-only at three model scales (1.5B, 3B, 7B).

Evidence: Ablations over teacher pre-training, retirement triggers, and α threshold sensitivity; comparisons vs. SFT, GRPO, continuous OPD, and SAGE-OPD.

The Big Picture

Long-horizon agentic tasks—booking a product on a website, completing a household chore sequence—require LLMs to make dozens of sequential decisions before receiving any feedback. Reinforcement learning can discover

effective strategies from this sparse reward but converges slowly because the credit assignment problem is severe: which of the 40 actions in a successful trajectory were actually critical?

On-policy distillation (OPD) offers a shortcut: a teacher that has privileged information (a task plan, a ground-truth action sequence, oracle environment state) generates dense token-level supervision, and the student imitates the teacher while simultaneously learning from RL. Prior work treats OPD as a fixed co-training objective. RetireOPD shows this is wrong: the benefit of distillation is not constant across training, and forcing the student to keep imitating a teacher it has already surpassed in some dimensions causes active regression.

Why Existing Approaches Fall Short

Pure RL (GRPO) trains with a clipped surrogate policy gradient but has no mechanism to bootstrap from the teacher’s privileged knowledge. The agent must discover strategies entirely from environment feedback, requiring many more episodes to converge.

Continuous joint OPD + RL combines distillation and RL throughout training. Two problems arise: (1) a teacher that was only marginally better than the student at initialisation, because privileged information alone does not make a teacher reliable in agentic settings where the teacher must also plan correctly, provides harmful early supervision; (2) later in training, the student explores behaviours beyond the teacher’s repertoire that OPD actively suppresses.

SAGE-OPD uses a predefined schedule to taper OPD linearly, which is better than constant OPD but cannot adapt to the variable rate at which different model scales or task complexities saturate distillation.

Core Mechanism

Phase 1: Teacher pre-training. A skill-conditioned teacher $\pi_T(a_t | s_t, z)$ is initialised from the base LLM and fine-tuned with GRPO on environment rewards augmented by a skill-following term that rewards use of the provided privilege z (e.g., sub-task labels, reference trajectories). This phase ensures the teacher is reliably better than a skill-free policy before distillation begins.

Phase 2: Student joint training. A skill-free student $\pi_S(a_t | s_t)$ is trained with:

$$\mathcal{L}_S = \mathcal{L}_{\text{GRPO}}(\pi_S) + \beta \cdot \mathcal{L}_{\text{OPD}}(\pi_T, \pi_S),$$

with teacher parameters frozen.

Adaptive Retirement. At each training step, two signals are monitored:

- *Discrepancy:* $\Delta_t = \text{KL}(\pi_T || \pi_S)$ averaged over recent rollouts.

- *Success gap*: $\rho_t = R_S^t/R_T$, the student’s rolling success rate relative to the teacher’s fixed rate.

Retirement fires when:

$$\Delta_t - \Delta_{t-k} < \epsilon_\Delta \quad \text{AND} \quad \rho_t \geq \alpha.$$

After retirement, $\beta=0$ and only $\mathcal{L}_{\text{GRPO}}$ is used.

Technical Formulation

The GRPO loss is:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}_\tau \sum_t \min \left(\frac{\pi_S(a_t)}{\pi_{\text{ref}}(a_t)} \hat{A}_t, \text{clip} \left(\frac{\pi_S}{\pi_{\text{ref}}}, 1 \pm \epsilon \right) \hat{A}_t \right)$$

where $\hat{A}_t$ is the group-relative advantage and π_{ref} is the reference (SFT checkpoint or retirement-point snapshot).

The OPD loss is:

$$\mathcal{L}_{\text{OPD}} = \mathbb{E}_\tau \sum_t \text{KL}(\pi_T(\cdot | s_t, z) \parallel \pi_S(\cdot | s_t)).$$

The teacher pre-training loss adds a skill-following reward:

$$r_T = r_{\text{env}} + \lambda_z \cdot \mathbf{1}[\pi_T \text{ uses } z].$$

Default hyperparameters: rolling window $k = 100$, convergence threshold $\epsilon_\Delta = 0.01$, success-fraction threshold $\alpha = 0.85$, distillation weight $\beta = 0.5$.

Learning or Inference Procedure

1. Pre-train skill-conditioned teacher with GRPO + skill reward for T_{teacher} steps until teacher success rate saturates.
2. Initialise student from base LLM (same checkpoint as teacher init).
3. Train student jointly with $\mathcal{L}_{\text{GRPO}} + \beta \mathcal{L}_{\text{OPD}}$.
4. At each step compute Δ_t and ρ_t ; fire retirement condition if met; set $\beta=0$ and continue RL-only.
5. At inference: deploy skill-free student π_S . Teacher and privilege z are not needed.

What the Guarantee Says

No formal guarantee is provided. The paper motivates retirement theoretically: under a noisy oracle model where the teacher is correct with probability $p > 0.5$ on each token, continuing OPD past the saturation point has *negative* expected alignment between the OPD gradient and the true reward gradient. Retirement is designed to detect this saturation empirically before the misalignment damages performance.

Experimental Findings

Evaluated on Qwen2.5 at three scales (1.5B, 3B, 7B):

Method	ALFWorld	WebShop
SFT	54.2	59.3
GRPO only	67.1	68.4
Continuous OPD+RL	71.3	72.1
SAGE-OPD	73.8	74.5
RetireOPD	85.9	87.4

Success rate (%), 7B scale. RetireOPD exceeds SAGE-OPD by 12.1% on ALFWorld and 12.9% on WebShop, confirming that adaptive retirement, not just scheduled tapering, is the key.

Ablations and Interpretation

No teacher pre-training: -8.3% on ALFWorld; the teacher provides unreliable supervision at initialisation, corrupting early student gradients.

No retirement (continuous OPD): -14.6% ; continued OPD past saturation actively suppresses productive student exploration.

Discrepancy-only retirement: Premature retirement at 1.5B scale (-4.1%); the student retires before reaching competence.

Success-gap-only retirement: Over-extended distillation at 7B scale (-2.7%); threshold is met late after OPD stops improving.

α sensitivity: $\alpha \in [0.80, 0.90]$ is robust; $\alpha < 0.70$ causes premature retirement; $\alpha > 0.95$ delays retirement past the benefit window.

Reference

Yan Yu, Zhengxi Lu, Yizhou Liu, Yichen Pan, Aozhe Wang, Qipeng Chen, Hua Yang, Wenqi Zhang, Weiming Lu, Qianglong Chen, Yongliang Shen. **RetireOPD: Self-Retiring On-Policy Distillation for Agentic Reinforcement Learning**. arXiv:2609.20784, September 2026. <https://arxiv.org/abs/2609.20784>